

Mathematics Bootcamp

General Equilibrium and the Social Planner

Miguel Vázquez-Carrero

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Today

- **Infinite paths:** recurrence, initial condition, and boundary at infinity
- **Household:** dated goods, shadow values, and the Euler equation
- **Competitive equilibrium:** definition, clearing, and characterization
- **Planner:** efficiency, supporting prices, and existence

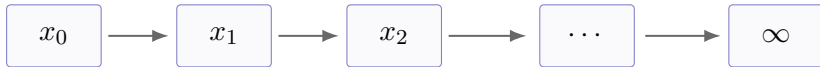
Day 4 Chose a Point; Day 5 Chooses an Infinite Path

Day 4 studied a scalar choice

$$x \in D \subseteq \mathbb{R}.$$

Today the choice is an infinite sequence:

$$z = (x_0, x_1, x_2, \dots).$$



In the one-state model below, a candidate path satisfies one Euler equation at every date and one transversality condition at infinity.

The Infinite Lagrangean Generates Candidate Conditions

Formally attach one multiplier to every dated constraint:

$$\max_{z=(z_t)_{t=0}^{\infty}} F(z) \quad \text{s.t.} \quad g_t(z) = 0, t = 0, 1, \dots, \quad L(z; \lambda) = F(z) + \sum_{t=0}^{\infty} \lambda_t g_t(z).$$

$$g_0(z) = 0$$

$$\lambda_0$$

$$g_1(z) = 0$$

$$\lambda_1$$

$$\dots$$
$$\dots$$

$$g_t(z) = 0$$

$$\lambda_t$$

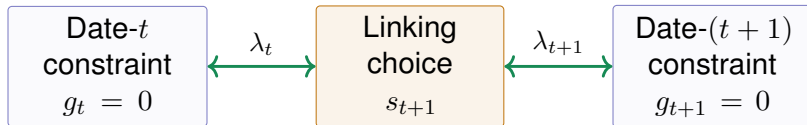
What Is Formal Here?

A general multiplier theorem for sequence spaces is beyond our scope. We use this expression to generate date-by-date candidates; feasibility, concavity, and a model-specific transversality condition do the verification.

A Linking Choice Creates a Recurrence

A variable carried from t to $t + 1$ appears in two neighboring constraints:

$$g_t(\dots, s_{t+1}) = 0, \quad g_{t+1}(s_{t+1}, \dots) = 0.$$



Differentiating with respect to s_{t+1} links adjacent multipliers. Eliminating them typically produces

$$\Phi(s_t, s_{t+1}, s_{t+2}) = 0.$$

This is a second-order difference equation.

A Recurrence Needs Conditions at Both Ends

A recurrence alone does not determine a path:

$$\Phi(s_t, s_{t+1}, s_{t+2}) = 0, \quad t = 0, 1, \dots$$

In the model below, a candidate path also needs an initial condition and a model-specific transversality condition:

$$s_0 \text{ given, } \quad \lim_{t \rightarrow \infty} \lambda_t s_{t+1} = 0.$$

Infinite-Horizon Logic

- (1) Existence requires an argument about feasible sequences and tails.
- (2) A strictly concave objective on a convex feasible set has at most one maximizer.
- (3) First-order conditions generate candidates. Feasibility, concavity, and the TVC perform verification.

The Abstract Recurrence Becomes an Euler Equation

Abstract mathematics

Dynamic economy

Sequence $z = (z_t)_{t=0}^{\infty}$

Paths of consumption, assets, and capital

Constraint $g_t(z) = 0$

Dated budget or resource constraint

Linking choice s_{t+1}

Saving or next-period capital

Multiplier λ_t

Shadow value of resources at date t

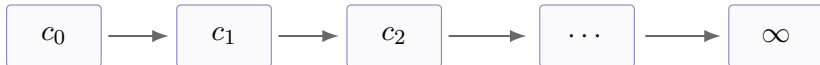
Recurrence

Euler equation

Dynamic economics applies constrained optimization to choices linked across dates.

A Date Label Creates a Different Commodity

A commodity is identified not only by its physical features, but also by its date and state. In this deterministic model, date is the key label:



Different Date, Different Good

Consumption at t and $t + 1$ are distinct commodities. Saving exchanges the current commodity for a future one; the interest rate will determine their relative price.

The Environment Is a List of Primitives

Dynamic Production Economy

The environment is

$$E = (\mathbb{N}_0, \beta, u, f, (A_t)_{t=0}^{\infty}, k_0) .$$

These primitives are fixed before prices and allocations are determined.

Primitives

- Dates $t = 0, 1, \dots$
- Preferences: β and u
- Technology: f and A_t

Initial Resources

- The household owns $a_0 = k_0 > 0$.
- Labor supply is one unit each date.
- Capital does not depreciate: $\delta = 0$.

The Transversality Condition Controls the Tail

The model uses dates $t = 0, 1, \dots$ and assumes:

- a deterministic closed economy with one representative price-taking household and firm;
- u is twice differentiable, $u' > 0$, $u'' < 0$, and $\lim_{c \downarrow 0} u'(c) = +\infty$;
- f is increasing, concave, and constant returns to scale; moreover, $k_0 = a_0 > 0$ and $\delta = 0$.

Transversality Condition

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) a_{t+1} = 0.$$

The Euler equation is local; this condition controls the tail.

The Household Ranks Entire Consumption Paths

Lifetime Utility

For $c = (c_0, c_1, c_2, \dots)$,

$$U(c) = \sum_{t=0}^{\infty} \beta^t u(c_t), \quad 0 < \beta < 1.$$

The discount factor β gives the relative weight on later utility. We assume that this sum is well-defined and finite on admissible paths.

Monotonicity

$$u'(c_t) > 0$$

More consumption is preferred.

Strict concavity

$$u''(c_t) < 0$$

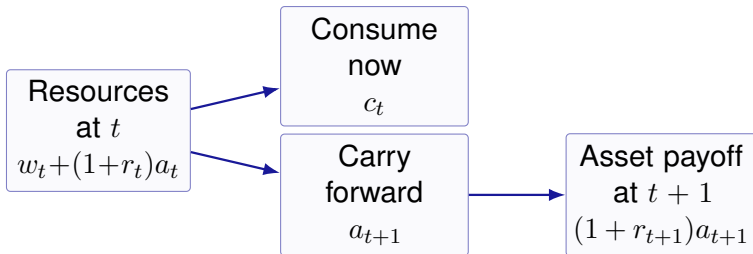
Marginal utility falls with consumption.

Saving Transfers Resources Across Dates

At date t , the household owns assets a_t , supplies one unit of labor, and receives wage and capital income.

Period Budget Constraint

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t.$$



The Household Chooses an Infinite Path at Given Prices

Given the price paths $(w_t, r_t)_{t=0}^{\infty}$ and initial assets a_0 , the household solves

$$\max_{(c_t, a_{t+1})_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t, \quad c_t \geq 0, \quad a_{t+1} \geq 0, \quad t = 0, 1, \dots,$$

a_0 given.

Admissibility and Price Taking

The restriction $a_{t+1} \geq 0$ is a no-borrowing constraint. It is sufficient to rule out Ponzi finance, but stronger than a no-Ponzi condition. The household treats every wage and interest rate as given.

One Shadow Value Is Needed for Every Date

At an interior path, write the formal Lagrangean

$$L = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t [w_t + (1 + r_t)a_t - c_t - a_{t+1}].$$

The first-order conditions imply

$$\beta^t u'(c_t) = \lambda_t, \quad t = 0, 1, \dots,$$

$$\lambda_t = (1 + r_{t+1})\lambda_{t+1}, \quad t = 0, 1, \dots$$

Interpretation

λ_t is the increase in lifetime utility from one additional unit of the consumption good available at date t .

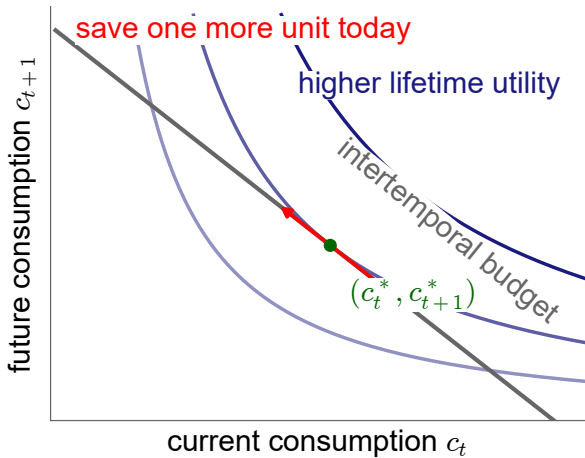
Tangency Across Dates Gives the Euler Equation

The household treats consumption at t and $t + 1$ as two different goods. At an interior optimum,

$$\frac{\beta u'(c_{t+1})}{u'(c_t)} = \frac{1}{1 + r_{t+1}}.$$

Same Mathematics as Day 4

The MRS equals the relative price. The only new feature is that the goods carry date labels.



The Euler Equation Balances Sacrifice and Return

Combine the two first-order conditions:

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).$$

Cost of Saving One More Unit

One unit is forgone today, reducing lifetime utility by

$$\beta^t u'(c_t).$$

Benefit One Date Later

Tomorrow it yields $1 + r_{t+1}$ units. Lifetime utility rises by

$$\beta^{t+1}(1 + r_{t+1})u'(c_{t+1}).$$

At an interior optimum, these two marginal effects are equal.

The Firm Transforms Capital and Labor into Output

In every date t , the firm rents capital k_t , hires labor ℓ_t , and produces the single final good:

$$y_t = A_t f(k_t, \ell_t) = A_t k_t^\alpha \ell_t^{1-\alpha}, \quad 0 < \alpha < 1.$$

Technology

A_t is total factor productivity. It scales output at every input pair; f remains the standard Cobb–Douglas function.

Constant Returns

Scaling both inputs by one factor scales output by that factor. Price taking then implies zero profits.

Throughout, $\delta = 0$: capital is fully preserved between dates.

The Firm's Problem Is Static Each Date

Normalize the final-good price to one. Given w_t and r_t , the firm solves

$$\max_{k_t, \ell_t \geq 0} \{A_t f(k_t, \ell_t) - w_t \ell_t - r_t k_t\}.$$

At a positive-input optimum,

$$w_t = A_t \frac{\partial f(k_t, \ell_t)}{\partial \ell_t} = A_t f_2(k_t, \ell_t), \quad r_t = A_t \frac{\partial f(k_t, \ell_t)}{\partial k_t} = A_t f_1(k_t, \ell_t).$$

Why the Firm Problem Is Static

This is the one-period problem from Day 4, repeated at every date. The firm rents inputs, produces, and sells output within date t ; the household owns the capital and chooses how much to carry into the next date.

Capital Earns r_t and Returns Intact

For Cobb–Douglas production,

$$w_t = (1 - \alpha)A_t k_t^\alpha \ell_t^{-\alpha}, \quad r_t = \alpha A_t k_t^{\alpha-1} \ell_t^{1-\alpha}.$$

The firm pays r_t for one unit of capital. Since capital is returned intact, the household receives the gross payoff $1 + r_t$.

Constant returns and the first-order conditions imply

$$A_t f(k_t, \ell_t) = w_t \ell_t + r_t k_t,$$

so competitive profits are zero.

Feasibility Restricts Every Date's Allocation

Allocation and Feasibility

A physical allocation is a sequence

$$z = (c_t, k_{t+1}, \ell_t)_{t=0}^{\infty}.$$

It is **feasible** if, for every $t = 0, 1, \dots$,

$$c_t \geq 0, \quad k_{t+1} \geq 0, \quad 0 \leq \ell_t \leq 1,$$

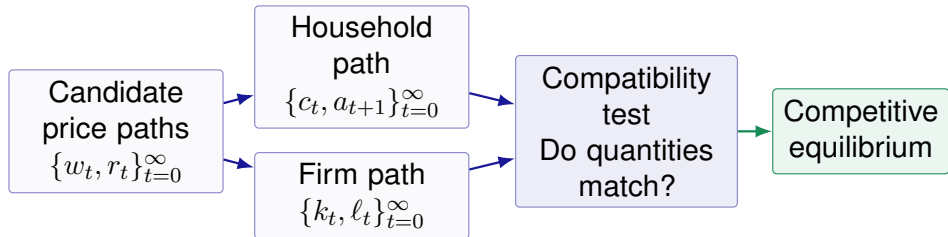
and

$$c_t + k_{t+1} \leq A_t k_t^{\alpha} \ell_t^{1-\alpha} + k_t.$$

Feasibility is physical and contains no prices. Monotonicity makes the resource inequality bind at an optimum.

General Equilibrium Makes Separate Choices Compatible

The household and firm have now solved separate optimization problems. General equilibrium asks whether their choices can hold at the same prices.



The Definition Is a Test, Not an Adjustment Story

It says which price-allocation paths qualify as equilibria. It does not say how an economy discovers those prices.

Optimal Choice May Be Set-Valued

Household Correspondence

At prices (w, r) , define

$$H(w, r) = \{\text{household-optimal paths}\}.$$

It may contain more than one path (c, a) .

Firm Correspondence

At prices (w_t, r_t) , define

$$F_t(w_t, r_t) = \{\text{optimal input pairs at } t\}.$$

It may contain more than one pair (k, ℓ) .

Under constant returns, F_t can be empty or set-valued. If one plan earns positive profit, scaling makes profit unbounded. At supporting zero-profit prices, many scales may be optimal. Equilibrium selects mutually compatible elements of both correspondences.

Definition and Characterization Answer Different Questions

DEFINITION

- Names the objects that constitute an equilibrium.
- States optimization and market clearing.
- Does not require derivatives or first-order conditions.

CHARACTERIZATION

- Derives equations every equilibrium must satisfy.
- Uses first-order conditions or KKT under their hypotheses.
- Needs extra conditions for sufficiency, existence, or uniqueness.

Equilibrium Adds Clearing to Optimal Choice

Competitive Equilibrium

Given a_0 , a competitive equilibrium is a collection

$$((c_t, a_{t+1}), (k_t, \ell_t), (w_t, r_t))_{t=0}^{\infty}$$

with $(w_t, r_t) \in \mathbb{R}_{++}^2$ for every t , such that:

1. $(c, a) \in H(w, r)$;
2. $(k_t, \ell_t) \in F_t(w_t, r_t)$ for every t ;
3. markets clear at every date:

$$a_t = k_t, \quad \ell_t = 1, \quad c_t + k_{t+1} = A_t f(k_t, \ell_t) + k_t.$$

Market Clearing Links Individual and Aggregate Quantities

Asset Market

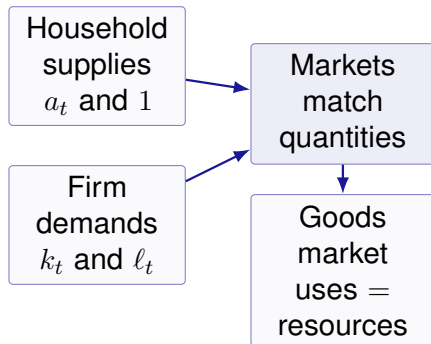
$$a_t = k_t.$$

Labor Market

$$\ell_t = 1.$$

Goods Market

$$c_t + k_{t+1} = A_t f(k_t, \ell_t) + k_t.$$



Walras' Law Makes One Clearing Condition Redundant

Suppose the household and firm optimize and the asset and labor markets clear.

(1) Substitute $a_t = k_t$ and $a_{t+1} = k_{t+1}$ into the household budget:

$$c_t + k_{t+1} = w_t + (1 + r_t)k_t.$$

(2) Subtract the capital carried into date t :

$$c_t + k_{t+1} - k_t = w_t + r_t k_t.$$

(3) Since $\ell_t = 1$ and constant returns imply zero profit,

$$w_t \ell_t + r_t k_t = A_t f(k_t, \ell_t).$$

(4) Therefore the goods market clears:

$$c_t + k_{t+1} = A_t f(k_t, \ell_t) + k_t.$$

Firm Optimality Makes Prices Functions of Capital

To characterize equilibrium, set $A_t = 1$ and use $\ell_t = 1$.

General Technology

$$w_t = f_\ell(k_t, 1),$$

$$r_t = f_k(k_t, 1).$$

Cobb–Douglas Technology

$$w_t = (1 - \alpha)k_t^\alpha,$$

$$r_t = \alpha k_t^{\alpha-1}.$$

Firm optimality and market clearing remove prices as independent unknowns.

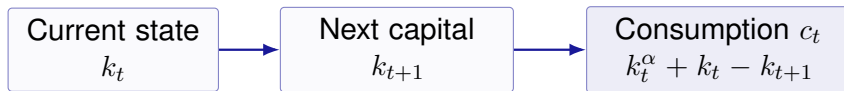
Feasibility Recovers Consumption from Capital

With $\ell_t = 1$ and $A_t = 1$, the resource constraint is

$$c_t + k_{t+1} = k_t^\alpha + k_t.$$

Hence,

$$c_t = k_t^\alpha + k_t - k_{t+1}.$$



Once the capital path is known, the allocation and prices can be recovered.

The Euler Equation Becomes a Recurrence in Capital

For $t = 0, 1, \dots$, substitute equilibrium prices and consumption into the household Euler equation:

$$\begin{aligned} u'(k_t^\alpha + k_t - k_{t+1}) \\ = \beta [1 + \alpha k_{t+1}^{\alpha-1}] u'(k_{t+1}^\alpha + k_{t+1} - k_{t+2}). \end{aligned}$$

This is a second-order difference equation in (k_t, k_{t+1}, k_{t+2}) . Combine it with

$$k_0 = a_0, \quad c_t \geq 0, \quad k_t \geq 0, \quad \lim_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} = 0.$$

A Characterization Is Not an Existence Theorem

Necessary Direction

interior equilibrium $\implies$ capital recurrence.

Converse Direction

capital recurrence + feasibility + initial condition
+ transversality + sufficient optimality conditions $\implies$ equilibrium.

Neither implication proves that a solution exists or that it is unique.

Efficiency Compares Only Feasible Allocations

Efficiency

A feasible allocation $z^* = (c_t^*, k_{t+1}^*, \ell_t^*)_{t=0}^{\infty}$ is efficient if there is no feasible $\tilde{z}$ such that $U(\tilde{c}) > U(c^*)$.

Efficiency uses only preferences and physical feasibility: no prices and no ownership rule. With one household, the planner selects the efficient path by maximizing the household's lifetime utility over feasible allocations.

The Planner Chooses the Best Feasible Path

With $A_t = 1$, the planner solves

$$\max_{(c_t, k_{t+1}, \ell_t)_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + k_{t+1} = k_t^{\alpha} \ell_t^{1-\alpha} + k_t, \quad t = 0, 1, \dots,$$

$$c_t \geq 0, \quad k_{t+1} \geq 0, \quad 0 \leq \ell_t \leq 1, \quad k_0 > 0 \text{ given.}$$

Since leisure does not enter utility and production increases in labor, $\ell_t = 1$. At an interior optimum,

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} = 0.$$

The Planner Has the Same Euler Equation

For the reduced problem, define

$$L = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t [k_t^\alpha + k_t - c_t - k_{t+1}].$$

The first-order conditions give

$$\lambda_t = \beta^t u'(c_t), \quad t = 0, 1, \dots,$$

$$\lambda_t = \lambda_{t+1} [\alpha k_{t+1}^{\alpha-1} + 1], \quad t = 0, 1, \dots$$

Therefore,

$$\boxed{u'(c_t) = \beta [\alpha k_{t+1}^{\alpha-1} + 1] u'(c_{t+1}).}$$

The Market and Planner Euler Equations Coincide

Competitive Equilibrium

Household Euler equation:

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).$$

Firm return and clearing:

$$1 + r_{t+1} = 1 + \alpha k_{t+1}^{\alpha-1}.$$

=

Social Planner

Planner return:

$$1 + \alpha k_{t+1}^{\alpha-1}.$$

Planner Euler equation:

$$u'(c_t) = \beta [1 + \alpha k_{t+1}^{\alpha-1}] u'(c_{t+1}).$$

The equality is economically powerful only because concavity and transversality make these equations sufficient.

Welfare Theorems Link Market and Planner

Assume lifetime utility is well-defined, the relevant optima are interior, the problems are concave, and both transversality conditions hold.

First Welfare Theorem

Every equilibrium allocation is a planner optimum:

equilibrium $\implies$ planner optimum.

Markets attain an efficient allocation.

Second Welfare Theorem

Every interior planner optimum can be supported by prices:

planner optimum $\implies$ equilibrium.

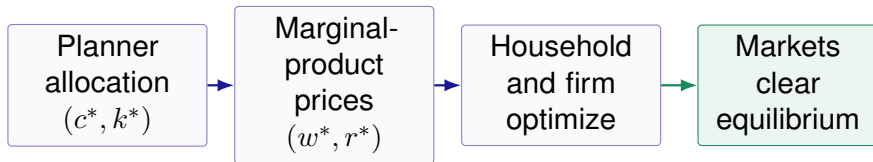
Marginal products construct the prices.

For an infinite horizon, these hypotheses are part of the theorem; the Euler equations alone are not sufficient.

Supporting Prices Decentralize the Planner Allocation

Start from an interior planner solution $(c_t^*, k_{t+1}^*)_{t=0}^\infty$ and define

$$w_t^* = (1 - \alpha)(k_t^*)^\alpha, \quad r_t^* = \alpha(k_t^*)^{\alpha-1}.$$



- (1) Concavity makes the firm's marginal-product conditions sufficient.
- (2) The planner Euler equation and transversality verify household optimality.
- (3) Set $a_t^* = k_t^*$ and $\ell_t^* = 1$; planner feasibility is goods clearing.

Infinite-Horizon Existence Requires a Separate Theorem

This is a problem over sequences, so the finite-dimensional Extreme Value Theorem does not apply directly. A rigorous existence proof must specify:

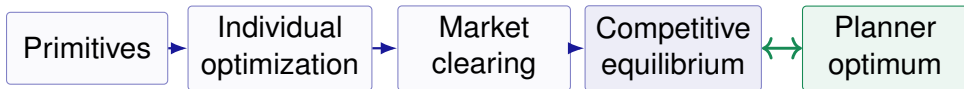
- (1) a topology on the path space;
- (2) compactness or sequential compactness of the feasible set;
- (3) continuity and uniform control of the discounted tail.

Two Separate Conclusions

planner solution exists + supporting-price theorem $\implies$ equilibrium exists,

planner solution is unique + First Welfare Theorem $\implies$ equilibrium allocation is unique.

Definition Comes First; Characterization Makes It Usable



General equilibrium is optimization made mutually consistent.